
Algorithm 1 Meta-evolve

Inputs: baseline; G_{total} total generations; G_{simplify} simplification generations;
 O offspring; P population size; N_{init} initial evaluation seeds;
 N_{reeval} reevaluation seeds; F fitness metric.

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1: population  $\leftarrow$  [baseline]
2: for  $i = 0, \dots, G_{\text{total}} - 1$  do
  # Generate offspring
3:   for  $o = 0, \dots, O - 1$  do
4:     sr_operator  $\leftarrow$  RANDOMCHOICE([mutation, survival, selection, loss])
5:     if  $i = 0$  then
6:       mode  $\leftarrow$  explore
7:     else if  $G_{\text{total}} - i < G_{\text{simplify}}$  then
8:       mode  $\leftarrow$  simplify
9:     else
10:      mode  $\leftarrow$  RANDOMCHOICE([explore, refine, crossover, simplify])
11:    end if
12:    parent  $\leftarrow$  TOURNAMENTSELECTION(population,  $n = 2$ )
13:    if mode = crossover then
14:      parent2  $\leftarrow$  TOURNAMENTSELECTION(population,  $n = 2$ )
15:    end if
16:    feedback  $\leftarrow$  GENERATETRACEFEEDBACK(tasks, parent)
17:    prompt  $\leftarrow$  GENERATEPROMPT(sr_operator, mode, parent, feedback, parent2)
18:    repeat
19:      offspring  $\leftarrow$  LLM_ensemble.sample().sample_completion(prompt)
20:    until PASSESVALIDATION(offspring)
21:  end for
  # Parallel evaluation of offspring and existing population
22:  evaluation_batch  $\leftarrow$   $\{x: N_{\text{init}} \mid x \in \text{offspring}\}$ 
   $\cup \{x: N_{\text{reeval}} - N_{\text{init}} \mid x \in \text{reeval\_individuals}\}$ 
23:  PARALLELEVALUATION(evaluation_batch, tasks,  $F$ )
  # Form archive of scores, averaged over seeds
24:  for  $x \in \text{offspring}$  do
25:    archive[ $x$ ]  $\leftarrow$   $\frac{1}{N_{\text{init}}} \sum_{\text{seeds, tasks}} F(x, \text{task}, \text{seed})$ 
26:  end for
27:  for  $x \in \text{reeval\_individuals}$  do
28:    archive[ $x$ ]  $\leftarrow$   $\frac{1}{N_{\text{reeval}}} \sum_{\text{seeds, tasks}} F(x, \text{task}, \text{seed})$ 
29:  end for
30:  if  $G_{\text{total}} - i < G_{\text{simplify}}$  then # Task-based population
31:    For each task, add the individual with highest average score on that task to population.
32:    Fill remaining slots (up to  $P$ ) with individuals selected by highest overall score.
33:  else # LOC versus score population
34:    Form the Pareto frontier of score versus LOC.
35:    Fill remaining slots (up to  $P$ ) with individuals selected by highest overall score.
36:  end if
  # Reevaluate surviving offspring for  $N_{\text{reeval}}$  seeds total
37:  reeval_individuals  $\leftarrow$   $\{x \in \text{offspring} \mid x \in \text{new\_population}\}$ 
38: end for
  # Final reevaluation to ameliorate winner's curse
39: for  $x \in \text{population}$  do
40:   final_eval[ $x$ ]  $\leftarrow$   $\frac{1}{N_{\text{reeval}}} \sum_{\text{seeds, tasks}} F(x, \text{task}, \text{seed})$ 
41: end for
42: best_individual  $\leftarrow$   $\arg \max_{x \in \text{population}} \text{final\_eval}[x]$ 
43: return final_eval, best_individual
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